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RESEARCH ARTICLE

An LCI Controller for Single-Phase Brushless Exciter Synchronous Motors With Extended Operating Range

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ABSTRACT This paper proposes a modified field-oriented control scheme for load commutated inverter-fed synchronous motors with single-phase brushless exciters. The core novelty is a fully analytical, real-time calculation of the inverter firing angle embedded directly into the control loop, eliminating the reliance on static margin angles, simplified speed estimations, empirical look-up tables or external hardware sensors. Specifically, the proposed scheme: 1) improves pulse generation accuracy through a load-synchronized firing scheme based on estimated stator fluxes; 2) effectively regulates rotor speed and stator flux magnitude using a simplified brushless exciter model; and 3) extends the speed and torque operating range by continuously adapting the firing angle to the instantaneous load current and commutation inductance without requiring offline calibration. Consequently, this analytical formulation minimizes unnecessary safety margins, reducing overall average exciter field and stator currents, allowing the extension of speed and torque operating range. Hardware-in-the-loop tests validate the robust dynamic response under a typical milling load profile.

INDEX TERMS Brushless excitation system, field-oriented control, flux observer, load commutated inverter, synchronous motor drive, synchronous motors, thyristor motor drives.

ACRONYMS

BE	Brushless exciter.
FOC	Field-oriented control.
HIL	Hardware in the loop.
LCI	Load commutated inverter.
LUT	Look-up table.
RTS	Real time simulation.
SCR	Silicon controlled rectifier.
SM	Synchronous motor.
SPh-BE	Single-phase brushless exciter.
SPh-BESM	Single-phase brushless excited sync. motor.

NOMENCLATURE

α	Inverter firing angle.
α_r	Rectifier firing angle.
δ	Torque angle.
γ	Margin angle.
μ	Overlap angle.
ϕ	Stator flux angle.
φ	Power factor angle.
$\boldsymbol{\psi}_{dqs}$	Stator flux vector in dq-coordinates.
$\boldsymbol{\psi}_{MTs}$	Stator flux vector in MT-coordinates.
$\boldsymbol{\psi}_{\alpha\beta s}$	Stator flux vector in $\alpha\beta$-coordinates.
$\mathbf{i}_{dqs}$	Stator current vector in dq-coordinates.
$\mathbf{i}_{MTs}$	Stator current vector in MT-coordinates.
$\mathbf{i}_{\alpha\beta s}$	Stator current vector in $\alpha\beta$-coordinates.
$\boldsymbol{e}_{\alpha\beta s}$	Stator back-EMF vector in $\alpha\beta$-coordinates.

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$i'_{fd_{\text{EXC}}}$	Single-phase brushless exciter field current.
$v'_{fd_{\text{EXC}}}$	Single-phase brushless exciter field voltage.
v'_{fd}	Synchronous motor field voltage.
i_{DC}	DC-link current.
L_c	Commutation inductance.
L_{DC}	DC-link choke.
I_s	Stator current magnitude.
Ψ_s	Stator flux magnitude.
Ψ_r	Equivalent rotor flux magnitude.
τ_e	Electromagnetic torque.
τ_L	Load torque.
t_q	Thyristor extinction time.
t_0	Thyristor extinction time offset.
ω_r	Rotor speed.
V_{LL}	Line-to-line voltage.

I. INTRODUCTION

Due to its unique characteristics and advantages, the synchronous motor (SM) has been one of the favorites in various industrial applications. Normally, a SM operates at a constant speed regardless of the load, which is normally synchronized with the frequency of the supply line. Furthermore, by simply adjusting the field excitation, it can be used for power factor correction in electrical power systems. Moreover, a SM can be used in variable-speed applications if controlled by a suitable power electronics module. Synchronous motors are composed of an AC armature (stator) and a DC excited field winding (rotor). The excitation can be obtained from either a static or a brushless supply. The former consists of a DC voltage source, sliprings, and brushes, similar to the usual excitation system of a DC machine. In the brushless case, the excitation power is generated in AC form by a generator (usually a synchronous machine as well), which is mounted on and rotates with the main machine shaft. This exciter operates as an 'inverted machine' with a rotating armature and a stationary field winding. The AC power produced by the brushless exciter is directly sent to a rectifier mounted on the shaft. The rectifier converts the induced AC voltage to DC, which is then supplied to the main field winding of the machine through the leads attached to the shaft [1]. The main advantage of the brushless exciter is the elimination of slip rings and brushes, improving the reliability of the SM.

There are two usual topologies for brushless exciters, namely, single-phase and three-phase (also called asynchronous exciter or rotary transformer). Compared to the *single-phase brushless exciter* (SPh-BE) case, the three-phase brushless exciter has the benefit of generating rotor flux at standstill, which allows the SM to start and operate at low speeds [2], [3], [4]. Consequently, three-phase brushless exciter has become the favorite among practitioners and has been implemented in a wide range of applications, highlighting the generating systems of brushless aircraft [2], [5]. In contrast, little attention has been paid to the single-phase topology and its implementation in applications with variable-speed controllers, where thresholds of the nominal values of the SM stator and SPh-BE currents limit the operating range of the SM in terms of rotor speed and load torque. However, we believe that the low cost and higher reliability of SPh-BE make it an attractive option for industrial applications.

In applications ranging from a few MWs to hundreds of MWs, an SM driven by a *load commutated inverter* (LCI) is preferred over any other variable speed machine [6], [7], [8]. The use of LCI-based drives in primary ball mills ensures smooth and controlled acceleration from partial to full speed, reduces motor maintenance requirements, allows flexible operation above or below nominal speed, and provides substantial pullout torque for demanding load conditions. [9].

Existing control methodologies for LCI drives generally fall into four established categories, each with notable limitations: (i) inverter fixed angle [10] and (ii) speed-based estimations [11], they both often ignore the dynamics of the load current and the switching inductance during transients, and fail to adapt to instantaneous electrical states; while (iii) empirical Look-Up Tables (LUTs) [12] demand extensive offline commissioning and remain highly vulnerable to thermal and magnetic parameter variations; and finally, (iv) hardware-based margin controls [13] rely on dedicated physical sensors to measure overlap angles, which increases system complexity and reduces the overall robustness. These schemes can increase the magnetizing current and the field current consumption, thus limiting the electromagnetic torque production and the minimum rotor speed. To clearly illustrate these differences, Table 1 summarizes the qualitative limitations of these existing commutation control methodologies.

In addition, most controllers usually consider a load-synchronized switching policy based on terminal voltages [14], which reduces pulse generation accuracy due to the unavoidable presence of commutation notches and spikes in those voltages produced by the thyristors natural commutation.

Recent research addresses traditional LCI limitations through hardware modifications and advanced control algorithms. Hardware-focused efforts explore multi-pulse topologies to improve startup capabilities [15], permanent magnet machine designs [16], and hybrid LCI-VSI setups for induction machines [17], or specialized applications like electromagnetic levitation [18]. Meanwhile, control-focused advancements include improved sliding-mode observers for sensorless position detection [19], and optimized speed control to enhance grid synchronization for synchronous condensers [20].

The SM model has been well established in the rotor (dq) reference frame, presuming a control structure relatively simple; however, in many SM drives, the MT-reference frame is favored over the dq-reference frame due to its ability to control the power factor and the potential for optimal sizing of the stator-side converter [6], [7], [8]. Therefore, in *field-oriented control* (FOC), the machine is typically modeled in the stator flux (MT) reference frame where the control

TABLE 1. Comparison of commutation control strategies for LCI drives.

Commutation control strategy	Hardware sensor requirement	Commissioning & calibration effort	Transient load robustness	Parameter variation immunity	Power factor & current efficiency
Fixed-angle	Low (standard FOC)	Low (static parameterization)	Poor (high fault risk)	Low (requires large buffers)	Low (limits active torque)
Speed-based estimation	Low (standard FOC)	Low (curve parameterization)	Medium (fails on fast steps)	Low (ignores inductance change)	Medium (sub-optimal margin)
Look-up tables (LUT)	Low (standard FOC)	High (extensive mapping)	High (adapts within map)	Poor (fails on thermal drift)	High (good within profile)
Hardware margin control	High (extra physical sensors)	Medium (sensor calibration)	High (direct tracking)	High (direct measurement)	High (minimizes reactive current)

is designed. In [11], the FOC design for an LCI drive for a salient-pole SM in *MT*-reference frame is described in detail, establishing the following benefits:

(i) In FOC, rotor speed and stator flux magnitude control are decoupled and managed through separate control loops.
(ii) For salient-pole motors, there is a significant cross-coupling between the torque and flux axes, which is compensated in FOC through a feedforward action.
(iii) The necessary over-excitation in the field circuit for load commutation can be accurately determined in real time as a function of DC-link current, rotor speed and power factor.

The present paper proposes a modified FOC controller for an LCI-based drive for the single-phase brushless exciter synchronous motor (SPh-BESM) (shown in Fig. 1) with the following benefits and characteristics:

(i) An alternative load synchronized switching policy based on estimated stator flux signals by an adaptive flux observer.
(ii) A simplification of the SPh-BE to consider its effect for control purposes.
(iii) A firing angle controller for the inverter side, which extends the speed and load torque operating ranges.

The extended speed range is highly appreciated in high-power ball mills in the mining industry. As steel grinding balls wear out over time, operators reduce the mill speed to extend their lifespan while maintaining grinding output. The proposed controller optimizes the commutation margin, allowing the drive to sustain high load torque or lower speeds without exceeding stator current limits. This allows steeper speed reductions than standard LCI controls, directly maximizing the lifespan of the grinding balls and reducing maintenance costs.

The rest of the paper is structured as follows: Section II describes the setup comprising a SM and LCI. Section III presents the conventional FOC controller in the *MT*-coordinates. Section IV outlines the design of the proposed controller modifications. Section V exhibits the experimental results based on hardware-in-the-loop (HIL) tests. Finally, Section VI provides concluding remarks of the work.

II. SYSTEM DESCRIPTION

This section introduces the driving system, starting with the synchronous machine equations (applicable to both SM and SPh-BESM [21]) in different coordinates according to their purpose. It also describes the LCI natural commutation of the system shown in Fig. 1.

A. SYNCHRONOUS MACHINE DYNAMICAL MODEL

This paper focuses specifically on the fluxes and currents of a salient-pole synchronous machine. Accordingly, the flux and current equations are derived in different reference frame coordinates for the following main purposes:

(i) *$\alpha\beta$-coordinates*. Necessary to simplify the three-phase AC sensor variables into an equivalent two-phase orthogonal system, providing the fundamental stationary framework required to compute the estimated stator flux magnitude $\hat{\Psi}_s$ and rotor flux magnitude $\hat{\Psi}_r$.
(ii) *dq-coordinates*. Aligning the frame with the rotor position is necessary to convert sinusoidal AC variables into DC quantities (during steady state). This allows to determine the estimated field current $\hat{i}_{fd}$ based on i_d and i_q out of the synchronous motor model.
(iii) *MT-coordinates*. The frame is oriented with the stator flux, which is essential for designing the FOC controllers and calculating the torque angle δ. Modeling in the *MT* frame is strictly necessary for LCI drives as it decouples torque and flux control actions, compensates for salient-pole cross-coupling terms and allows the on-line determination of the exact field over-excitation required for a safe load commutation.

1) ROTOR FLUX MAGNITUDE AND FIELD VOLTAGE DEFINITIONS FOR SM AND SPH-BE IN *DQ*-COORDINATES

The stator flux vector in *dq*-coordinates is defined as

$$\boldsymbol{\psi}_{dqs} = \begin{bmatrix} \psi_{ds} \\ \psi_{qs} \end{bmatrix} = \begin{bmatrix} L_d & 0 & L_{md} \\ 0 & L_q & 0 \end{bmatrix} \begin{bmatrix} i_{ds} \\ i_{qs} \\ i'_{fd} \end{bmatrix}, \tag{1}$$

where formulating the stator current components in *dq*-coordinates, i_{ds} and i_{qs}, yields the stator current vector $\mathbf{i}_{dqs}=[i_{ds},\ i_{qs}]^T$.

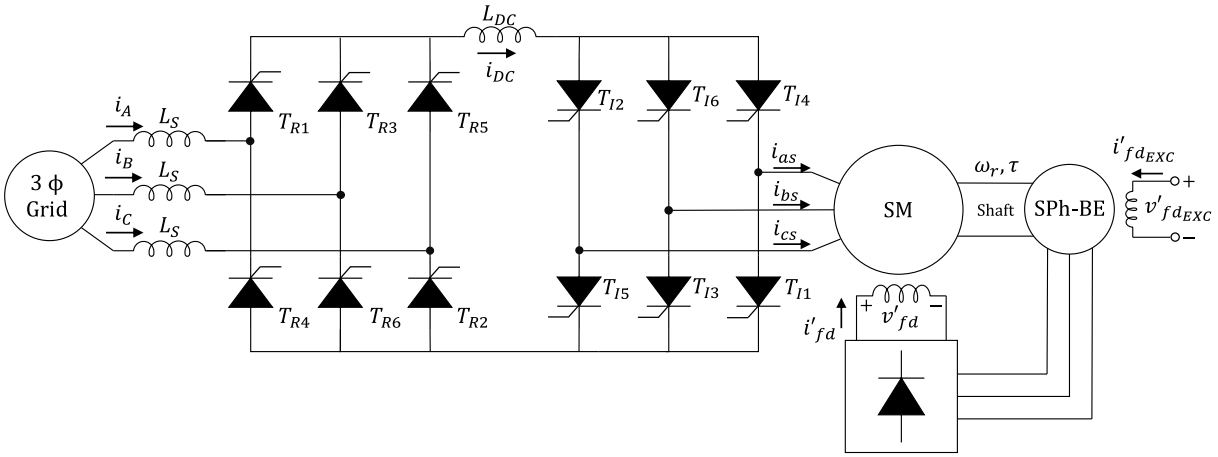


FIGURE 1. Circuit diagram of an LCI-based drive for a SPh-BESM.

The self-inductances in (1) comprise leakage and magnetizing inductances, that is, $L_d{=}L_{ls}+L_{md}$ and $L_q{=}L_{ls}+L_{mq}$; and the rotor side quantities have been referred to the stator side.

Furthermore, the electromagnetic torque is given by

$$\tau_e = \frac{3}{2}P\left(\psi_{ds}i_{qs} - \psi_{qs}i_{ds}\right), \tag{2}$$

where P is the number of pole-pairs in the rotor.

Substituting, in (2), ψ_{ds} and ψ_{qs}, as defined in (1), yields

$$\tau_e = \frac{3}{2}P\left[(L_d - L_q)i_{ds} + L_{md}i'_{fd}\right]i_{qs} \tag{3}$$

where the term inside the brackets on the right-hand-side (RHS) of (3) corresponds to the rotor flux magnitude Ψ_r in dq-coordinates [22]

$$\Psi_r \triangleq (L_d - L_q)i_{ds} + L_{md}i'_{fd}, \tag{4}$$

Additionally, the field voltage on the SPh-BE side $v'_{fd_{\rm EXC}}$ is defined by

$$v'_{fd_{\rm EXC}} = r'_{fd_{\rm EXC}}i'_{fd_{\rm EXC}} + \dot{\Psi}_{fd_{\rm EXC}}, \tag{5}$$

where $i'_{fd_{\rm EXC}}$ and $\Psi_{fd_{\rm EXC}}$ are the field current and flux on the SP-BE side; and the field voltage on the SM side v'_{fd} by

$$v'_{fd} = r'_{fd}i'_{fd} + \dot{\Psi}_{fd}, \tag{6}$$

where i'_{fd} and Ψ_{fd} are the field current and flux on the SM side. Due to its construction, v'_{fd} and i'_{fd} are inaccessible to be measured.

2) FLUX EQUATIONS IN *MT*-COORDINATES

The stator flux vector in MT-coordinates is defined in (7), as shown at the bottom of the page, where formulating the stator current components in MT-coordinates, i_{Ms} and i_{Ts}, yields the stator current vector $\mathbf{i}_{MTs}{=}[i_{Ms},\ i_{Ts}]^T$.

In [22], (7) was derived as a particular case of a description in an arbitrary XY-reference frame, where the torque angle δ was considered. Notice that the current flowing in the T-axis produces a flux linkage along the M-axis and vice-versa. This occurs due to the effect of saliency and must be considered for control design purposes [11].

Assuming that the transformation to the MT-reference frame is correctly performed, the stator flux magnitude in MT-coordinates is given by

$$\Psi_s = \left(L_{ls}{+}L_{md}\cos^2\delta{+}L_{mq}\sin^2\delta\right)i_{Ms}{+} + \left(\frac{L_{mq}{-}L_{md}}{2}\sin 2\delta\right)i_{Ts}{+}L_{md}\cos\delta\, i'_{fd}, \tag{8}$$

where $\Psi_s \triangleq |\boldsymbol{\psi}_{MTs}|$.

Grouping terms in (8), the following relationship is satisfied at every moment:

$$\Psi_s \triangleq \Psi_{AR} + \Psi_r, \tag{9}$$

where

$$\Psi_{AR} \triangleq \left(L_{ls} + L_{md}\cos^2\delta + L_{mq}\sin^2\delta\right)i_{Ms}{+} \tag{10}$$

$$+\left(\frac{L_{mq} - L_{md}}{2}\sin\delta\right)i_{Ts}, \tag{11}$$

is the armature reaction component which is the flux produced by i_{Ts} and i_{Ms} that magnetizes or demagnetizes $|\boldsymbol{\psi}_{MTs}|$,

$$\boldsymbol{\psi}_{MTs} = \begin{bmatrix}\psi_{Ms}\\ \psi_{Ts}\end{bmatrix} = \begin{bmatrix} L_{ls}{+}L_{md}\cos^2\delta{+}L_{mq}\sin^2\delta & \frac{L_{mq}{-}L_{md}}{2}\sin 2\delta & L_{md}\cos\delta \\ \frac{L_{mq}{-}L_{md}}{2}\sin 2\delta & L_{ls}{+}L_{mq}\cos^2\delta{+}L_{md}\sin^2\delta & -L_{md}\sin\delta \end{bmatrix}\begin{bmatrix} i_{Ms}\\ i_{Ts}\\ i'_{fd}\end{bmatrix} \tag{7}$$

and

$$\Psi_r \triangleq L_{md} \cos \delta i'_{fd}. \tag{12}$$

is the rotor flux magnitude in MT-coordinates.

3) STATOR AND ROTOR FLUX VECTORS IN $\alpha\beta$-COORDINATES

The dynamics of the stator flux vector in $\alpha\beta$-coordinates is given by

$$\dot{\boldsymbol{\psi}}_{\alpha\beta s} = \mathbf{e}_s = \mathbf{v}_T - r_s \mathbf{i}_{\alpha\beta s}, \tag{13}$$

where $\boldsymbol{\psi}_{\alpha\beta s} = [\psi_{\alpha s},\ \psi_{\beta s}]^T$ is the stator flux vector, $\mathbf{i}_{\alpha\beta s} = [i_{\alpha s},\ i_{\beta s}]^T$ is the stator current vector, $\mathbf{e}_s = [e_{\alpha s},\ e_{\beta s}]^T$ is the stator voltage vector, $\mathbf{v}_T = [v_{\alpha s},\ v_{\beta s}]^T$ is the terminal voltage vector, and r_s is the stator resistance.

Additionally, the rotor flux vector is given by

$$\boldsymbol{\psi}_{\alpha\beta r} = \boldsymbol{\psi}_{\alpha\beta s} - L_q \mathbf{i}_{\alpha\beta s}. \tag{14}$$

Remark 1: Since the transformations between frames MT, dq and $\alpha\beta$-reference frames are unitary rotations, then the following holds for the stator flux magnitude Ψ_s:

$$|\boldsymbol{\psi}_{MTs}| = |\boldsymbol{\psi}_{dqs}| = |\boldsymbol{\psi}_{\alpha\beta s}| = \Psi_s, \tag{15}$$

B. DESCRIPTION OF THE LCI NATURAL COMMUTATION

An LCI is a drive for AC machines based on a current source. In this system, AC currents are rectified by a silicon controlled rectifier (SCR) system, resulting in a DC current whose high ripple is filtered by a choke. This filtered DC current is fed to a SCR-based inverter that produces an AC current at the machine frequency with adjustable magnitude and phase [8]. The advantages of this approach are the following:

1) SCRs are highly reliable and capable of driving systems in the order of hundreds of MWs. Therefore, the combination of SM and LCI is the preferred choice in most high-power industrial applications [6], [7], [8].
2) The machine speed is controlled by regulating the torque produced by the motor. As the machine is fed with currents, the torque is directly proportional to the DC-link current, which facilitates the control design.
3) The motor power factor is established by adjusting the inverter firing angle α.

This paper is particularly interested in the study of the inverter commutation process, where the signals involved are shown in Fig. 2. The angle μ, corresponding to the notch in the line-to-line voltage signals, is called the overlap angle, whose expression was derived in [23]. The overlap angle μ measures the overlap of two phase currents during commutation, i.e., during the transfer of the DC-link current from one phase to another, and is given by

$$\mu = \arccos\left(\cos\alpha - 2\omega_r L_c i_{\mathrm{DC}}/V_{\mathrm{LL}}\right) - \alpha. \tag{16}$$

Provided that the SM operates at constant flux, then the term $2\omega_r L_c/V_{\mathrm{LL}}$ on RHS of (16) can be considered equal to a constant k_{Lc}, that is,

$$k_{Lc} = 2\omega_r L_c/V_{\mathrm{LL}}, \tag{17}$$

where ω_r is the electrical rotor speed, L_c is the commutation inductance and V_{LL} is the line-to-line voltage.

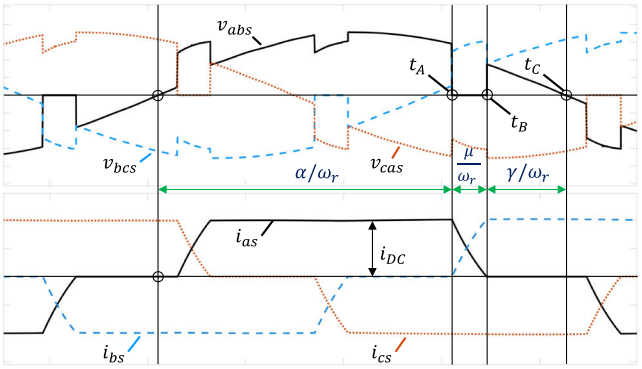


FIGURE 2. Line-to-line voltages (top) and phase currents (bottom) of a three-phase LCI-driven synchronous motor.

Moreover, Fig. 2 shows that the margin angle γ is the equivalent in radians to the time interval $[t_B,\ t_C]$, that is, between the return to zero of i_a and the zero crossing of v_{ab} [23]. This angle is given by

$$\gamma = \pi - \arccos\left(\cos\alpha - 2\omega_r L_c i_{\mathrm{DC}}/V_{\mathrm{LL}}\right). \tag{18}$$

Notice that, (18) can be simplified based on (17) as follows:

$$\gamma = \arccos\left(k_{Lc} i_{\mathrm{DC}} - \cos\alpha\right). \tag{19}$$

Now, to guarantee safe commutation, the following condition must be satisfied:

$$\gamma/\omega_r \geq t_q, \tag{20}$$

which holds for an α sufficiently small, where t_q is the thyristor extinction time. That is, t_q is the minimum time the thyristor T must be exposed to a sufficient reverse (negative) voltage to ensure the thyristor is turned off; otherwise, the component returns into a conduction state and a commutation failure may occur leading to a drive trip [23].

The power factor angle can be defined in terms of the firing and overlap angles as follows:

$$\varphi = \arccos\left(-\cos\left(\alpha + \mu/2\right)\right). \tag{21}$$

It should be noted that i_{Ms} is strictly related to α through both φ and i_{Ts}, where the latter is imposed by the load. Therefore, when the inverter firing angle is set to guarantee safe commutation according to (20), a suitable i_{Ms} is inherently delivering reactive power to commute the thyristors and demagnetize the stator flux, leading to a capacitive operation of the SM.

III. CONVENTIONAL FOC

A. TORQUE CONTROL

The speed controller regulates the rotor speed ω_r, while the current controller regulates the DC-link current i_{DC}. Both controllers use PI structures. The DC-link current flows through the inductor between the rectifier and the inverter of the LCI. The electromechanical torque τ_e produced by the SM is directly proportional to i_{DC}, i.e.,

$$\tau_e = k_\tau i_{\mathrm{DC}}, \tag{22}$$

where k_T is a proportional constant.

The proposed controller in [11] is performed with two PI regulators, one delivering the electromagnetic torque reference τ_e^* and the other one used to calculate the rectifier firing angle α_r, that is,

$$\tau_e^* = k_{p\omega}\tilde{\omega}_r + \frac{k_{p\omega}}{T_{i\omega}}\int \tilde{\omega}_r\, dt, \tag{23}$$

$$\alpha_r = \arccos(u_\alpha), \tag{24}$$

$$u_\alpha = k_{p\alpha}\tilde{I}_{\mathrm{DC}} + \frac{k_{p\alpha}}{T_{i\alpha}}\int \tilde{I}_{\mathrm{DC}}\, dt. \tag{25}$$

where $k_{p\omega}$ and $k_{p\alpha}$ are the proportional gains, $T_{i\omega}$ and $T_{i\alpha}$ are the integral time constants for the rotor speed and current controllers, respectively; $\tilde{\omega}_r \triangleq \omega_r^* - \omega_r$ is the rotor speed error with ω_r^* being the reference; and $\tilde{i}_{\mathrm{DC}} \triangleq i_{\mathrm{DC}}^* - i_{\mathrm{DC}}$ is the DC-link current error with i_{DC}^* being the reference given by

$$i_{\mathrm{DC}}^* = \frac{\pi}{2\sqrt{3}}\frac{\tau_e^*}{\hat{\Psi}_s \cos\varphi}. \tag{26}$$

The LCI-based drive makes the nominal 50 Hz or 60 Hz synchronous speed associated with AC power meaningless, and thus the upper speed limit is determined by mechanical limitations. In other words, the supply line and machine frequencies are decoupled from each other. [24].

B. STATOR FLUX MAGNITUDE CONTROL

The estimated stator flux magnitude is also regulated using a PI controller whose output is then used in the calculation of i_μ^*. The latter is the equivalent magnetizing current responsible for producing the stator flux.

1) DECOUPLING BETWEEN TORQUE AND FLUX CONTROL

The second term on the RHS of the stator flux magnitude (8) given by

$$g(\delta,\ i_{Ts}) = \left(\frac{L_{mq} - L_{md}}{2}\sin 2\delta\right) i_{Ts} \tag{27}$$

is referred to as the coupling term since the current flowing in the T-axis induces a flux linkage along the M-axis, in other words, the torque produced by i_{Ts} turns out to be coupled with the stator flux aligned to the M-axis. To deal with such coupling term, [11] proposed the following controller that delivers an expression for i_μ^*, which cancels out the coupling term through a feedforward action[1]:

$$i_\mu^* = \frac{1}{L_{md}}\left[u_\mu - \left(\frac{L_{mq} - L_{md}}{2}\sin 2\hat{\delta}\right) i_{Ts}\right], \tag{28}$$

$$u_\mu = k_{p\psi}\tilde{\Psi}_s + \frac{k_{p\psi}}{T_{i\psi}}\int \tilde{\Psi}_s\, dt. \tag{29}$$

where $k_{p\psi}$ is the proportional gain and $T_{i\psi}$ is the integral time constant; $\tilde{\Psi}_s \triangleq \Psi_s^* - \hat{\Psi}_s$ is the stator flux magnitude error, with Ψ_s^* being the stator flux magnitude reference.

2) OVER-EXCITATION OF THE FIELD CIRCUIT TO ALLOW NATURAL LOAD-COMMUTATION

The LCI natural commutation is possible after guaranteeing a leading power factor on the motor side. For this, the field current must be incremented by the following amount to overexcite the field circuit:

$$\Delta i_{fd}'^* = -\frac{\left(L_{ls} + L_{md}\cos^2\hat{\delta} + L_{mq}\sin^2\hat{\delta}\right) i_{Ms}}{L_{md}\cos\hat{\delta}}. \tag{30}$$

In [11], the following controller is proposed to regulate the field current (estimate) $\hat{i}_{fd}'$ that provides a reference v_{fd}' for the field voltage:

$$v_{fd}' = k_{pv}\tilde{i}_{fd}' + \frac{k_{pv}}{T_{iv}}\int \tilde{i}_{fd}'\, dt, \tag{31}$$

where k_{pv} is the proportional gain; T_{iv} is the integral time constant; $\tilde{i}_{fd}' \triangleq (i_{fd}'^* + \Delta i_{fd}'^*) - \hat{i}_{fd}'$ is the field current error with $\hat{i}_{fd}'$ to be defined later and $i_{fd}'^*$ being the SM field current reference given by

$$i_{fd}'^* = i_\mu^*/\cos\hat{\delta}, \tag{32}$$

where $\hat{\delta}$ is given by

$$\hat{\delta} = \arctan\frac{L_q i_{Ts}}{\hat{\Psi}_s - L_q i_{Ms}}. \tag{33}$$

Notice that reference $i_{fd}'^*$ has to be augmented by $\Delta i_{fd}'^*$ (30) to create a new reference given by $(i_{fd}'^* + \Delta i_{fd}'^*)$ used in $\tilde{i}_{fd}' \triangleq (i_{fd}'^* + \Delta i_{fd}'^*) - \hat{i}_{fd}'$ to guarantee the overexcitation of the field, as previously mentioned.

C. STATOR FLUX MAGNITUDE ESTIMATION

Flux estimation is an important task for the FOC implementation in high-performance drives for SM. The conventional estimation method, defined in $\alpha\beta$-coordinates, is based on an open-loop integration of the stator voltage vector $\mathbf{e}_{\alpha\beta s}$ (13), that is,

$$\hat{\boldsymbol{\psi}}_{\alpha\beta s} = \int \mathbf{e}_s\, dt = \int (\mathbf{v}_T - r_s\mathbf{i}_s)\, dt, \tag{34}$$

where $\hat{\boldsymbol{\psi}}_{\alpha\beta s}$ is the estimated $\alpha\beta$-stator flux vector.

[1]It is worth noting that other control schemes previously reported did not deal with the coupling between M and T, whose effect resulted in a non-smooth SM operation under FOC.

According to the cartesian-to-polar transformation, the estimated stator flux magnitude and angle are given by

$$\hat{\Psi}_s = \sqrt{\hat{\psi}_{\alpha s}^2 + \hat{\psi}_{\beta s}^2} \quad \hat{\phi} = \arctan\left(\hat{\psi}_{\beta s}/\hat{\psi}_{\alpha s}\right). \tag{35}$$

D. FIELD CURRENT ESTIMATION

Based on the definition of the stator flux vector in dq-coordinates, (3), the following current estimator, neglecting the damping windings, was proposed in [22]:

$$\hat{i}'_{fd} = \frac{\hat{\Psi}_r - (L_d - L_q) i_{ds}}{L_{md}}, \tag{36}$$

where the estimated rotor flux magnitude $\hat{\Psi}_r$ is obtained using (14), where the stator flux $\boldsymbol{\psi}_{\alpha\beta s}$ is replaced by its estimate $\hat{\boldsymbol{\psi}}_{\alpha\beta s}$ obtained with the flux observer algorithm (34), that is,

$$\hat{\Psi}_r = |\hat{\boldsymbol{\psi}}_{\alpha\beta r}| = |\hat{\boldsymbol{\psi}}_{\alpha\beta s} - L_q \mathbf{i}_{\alpha\beta s}|. \tag{37}$$

IV. PROPOSED α-MODIFIED FOC

Section III introduced the conventional FOC of an LCI-based drive for a salient-pole SM. In this section, the following three modifications are incorporated to enable the conventional FOC to handle an LCI-based drive for a SPh-BESM:

(i) An alternative switching policy based on estimated stator fluxes using an adaptive flux observer.
(ii) The consideration, in the control design, of a simplified model of the SPh-BE.
(iii) An algorithm for online calculation of the inverter firing angle.

The proposed control scheme, including these modifications, is hereafter referred to as the *α-modified FOC*.

The *new objectives* to pursue are

- Improve the pulse generation accuracy on the inverter-side.
- Regulate ω_r and $\hat{\Psi}_s$ for a SPh-BESM fed by an LCI-based drive.
- Extend the operating range of ω_r and τ_L by controlling α, while ensuring safe commutation.

A. FLUX-BASED LOAD-SYNCHRONIZED SWITCHING POLICY

The idea behind the alternative switching policy consists in implementing the synchronization stage using the *abc*-estimated stator fluxes as a cleaner and more suitable reference to guarantee proper switching of the SCRs in the LCI-based drive. Due to their orthogonality, these stator fluxes are in phase with terminal voltages (neglecting the resistance voltage drop), that is, $\hat{\psi}_{as}$ is in phase with v_{bcs}, $\hat{\psi}_{bs}$ is in phase with v_{cas} and $\hat{\psi}_{cs}$ is in phase with v_{abs}.

1) STATOR FLUX MAGNITUDE ESTIMATION

Stator fluxes and angle are estimated using the adaptive flux observer proposed in [25] and illustrated in Fig. 3. The observer considers variables expressed in $\alpha\beta$-coordinates,

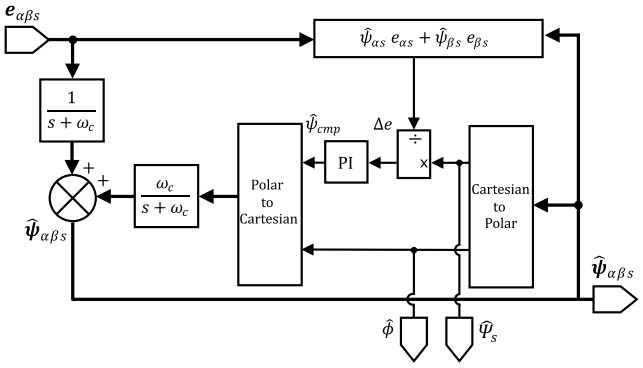


FIGURE 3. Block diagram of the adaptive flux observer [25].

and then the estimated fluxes are transformed back into *abc*-coordinates, resulting in $\boldsymbol{\psi}_{abcs}$. The process also considers transformations between cartesian and polar coordinates. It is important to note that the scheme inherently involves a filtering process. Therefore, estimated fluxes become cleaner signals than terminal voltages, as stated above. In particular, estimated fluxes are free of commutation notches and spikes, in contrast to terminal voltages shown in Fig. 2. Consequently, the use of estimated fluxes results in a more accurate pulse generation.

Conventional FOC estimates $\boldsymbol{\psi}_{\alpha\beta s}$ using (34); however, open-loop integration suffers from DC drift and initial-value errors. The adaptive flux observer overcomes these issues through adaptive compensation that takes advantage of the orthogonality of $\boldsymbol{\psi}_{\alpha\beta s}$ to $\boldsymbol{e}_{\alpha\beta s}$. As a result, the flux estimation remains accurate across different flux levels. As will become clear later, for flux control purposes, the magnitude $\hat{\Psi}_s$ serves as a feedback input. Additionally, variables that must be expressed in MT-coordinates can be obtained with the Park transformation using the estimated angle $\hat{\phi}$.

2) SWITCHING POLICY

A new switching policy to trigger the inverter-side is presented in Table 2, which is based on the vector of *abc*-estimated stator fluxes $\hat{\boldsymbol{\psi}}_{abcs}$=$[\hat{\psi}_{as},\ \hat{\psi}_{bs},\ \hat{\psi}_{cs}]^T$ for synchronization purposes. Notice that this is an alternative method to the switching policy presented in [26], where the authors used a phase-locked loop with zero-crossing detector and a flux observer for one flux level only.

TABLE 2. Load-synchronized switching policy based on synchronizing estimated stator fluxes.

Voltage condition			Triggered SCR
$(\hat{\psi}_{bs} < 0)$	$\wedge$	$(\hat{\psi}_{as} < 0)$	T_{I1}
$(\hat{\psi}_{as} > 0)$	$\wedge$	$(\hat{\psi}_{cs} > 0)$	T_{I2}
$(\hat{\psi}_{cs} < 0)$	$\wedge$	$(\hat{\psi}_{bs} < 0)$	T_{I3}
$(\hat{\psi}_{bs} > 0)$	$\wedge$	$(\hat{\psi}_{as} > 0)$	T_{I4}
$(\hat{\psi}_{as} < 0)$	$\wedge$	$(\hat{\psi}_{cs} < 0)$	T_{I5}
$(\hat{\psi}_{cs} > 0)$	$\wedge$	$(\hat{\psi}_{bs} > 0)$	T_{I6}

It is important to clarify that the most relevant information for synchronization purposes is the phase of the estimated stator fluxes, not necessarily their amplitudes. In addition, SCRs are turned off by natural commutation; thus, no turn-off signal is required.

B. SINGLE-PHASE BRUSHLESS EXCITER MODEL

To consider the SPh-BE in the control design process, it is required to derive an expression relating v'_{fd} on the SM side (5), and $v'_{fd_{\mathrm{EXC}}}$ on the SPh-BE side (6). Recall that, in general, the more $v'_{fd_{\mathrm{EXC}}}$, the more current $i'_{fd_{\mathrm{EXC}}}$ inducing a higher v'_{fd} on the SM side. Based on this, and on the fact that the SPh-BESM shown in Fig. 1 comprises a complex non-linear model, the following simplified model of both the SPh-BE and SM field winding (including its rectifier) based on [27] is proposed:

$$v'_{fd_{\mathrm{EXC}}} = k_{\mathrm{EXC}} v'_{fd}, \tag{38}$$

where v'_{fd} is also referred to as the rectified voltage in steady-state at no load-conditions. Expression (38) is used to simplify the field current PI tuning and thus speed up the drive commissioning. The constant k_{EXC} was identified via an open-loop, no-load voltage ramp (0.75–1.25 pu). Tests across the 0.8–1.2 pu speed range confirmed a speed-invariant slope, yielding $k_{EXC} = 0.118$. While this provides a rapid first-order approximation, the closed-loop PI regulator naturally absorbs any unmodeled non-linearities or dynamic transients under actual load conditions. Additionally, the simplified model has been expressed in terms of voltage variables to ensure direct compatibility with the stator flux controller introduced in the previous section.

Expression (38) involving k_{EXC} represents in fact a structural need. Omitting this parameter results in a sluggish step response, whereas considering it using a nominal value for k_{EXC} yields the optimal intended dynamics. The closed-loop system is highly robust to operating parameter drift, absorbing estimation errors of up to $\pm 30\%$ with stability. However, accurately identifying the nominal k_{EXC} remains vital, as overestimating it ($+30\%$) leads to excessive transient field voltages that may cause machine insulation damage.

C. PROPOSED INVERTER FIRING ANGLE CONTROL

1) EFFECT OF THE SPH-BE FIELD CURRENT AND STATOR CURRENT MAGNITUDE ON ω_R AND τ_L

Considering that the SPh-BESM is already in steady-state operation, a reduction in ω_r causes a reduction in the induced voltage E_A in the SM. This effect is captured by the following expression:

$$E_A = k \Psi_r \omega_r. \tag{39}$$

Besides, in general, the more excitation current $i'_{fd_{\mathrm{EXC}}}$ circulates, the more rotor flux Ψ_r is produced. Therefore, to keep E_A invariant, any reduction in ω_r can be compensated by increasing the current $i'_{fd_{EXC}}$. That is, the larger $i'_{fd_{\mathrm{EXC}}}$, the smaller ω_r. However, a maximum limit allowed $i'_{fd_{EXC}}$ limits the minimum reachable ω_r.

The expression of τ_e, under steady-state operation, disregarding mechanical losses ($\tau_e \approx \tau_L$), is given by

$$\tau_e = \frac{V_T E_A}{X_S} \sin \delta, \tag{40}$$

If V_T and X_S are constants, then an increment in τ_L corresponds to an increment in $E_A \sin \delta$. In particular, an increment in E_A can be achieved, as above discussed for (39), through an increment in $i'_{fd_{EXC}}$. In other words, compensation of larger τ_L can be achieved by increasing $i'_{fd_{\mathrm{EXC}}}$. Furthermore, τ_e is also a function of $I_s = |\mathbf{i}_{MTs}|$, and thus applying a greater τ_L demands higher I_s. Therefore, compensation of larger τ_L can be achieved by increasing both $i'_{fd_{\mathrm{EXC}}}$ and I_s.

Summarizing, the minimum and maximum thresholds of ω_r and τ_L that can be reached are actually restricted by $i'_{fd_{EXC}}$ and I_s. In what follows, a strategy is proposed to extend the operation of the SPh-BESM beyond the typical limits on ω_r and τ_L. For this, it is important to identify the available control input on the LCI driving the SPh-BESM that can be used to pursue this objective among the following 3 control input variables:

(i) The rectifier firing angle α_r, which is imposed by the current controller (24).
(ii) The SPh-BE field voltage $v'_{fd_{\mathrm{EXC}}}$, which is imposed by the stator flux controller (38).
(iii) The inverter firing angle α, which until now was selected to ensure only safe commutation.

However, α_r and $v'_{fd_{\mathrm{EXC}}}$ are not longer available as they are already imposed by other control loops, and thus α is the only remaining control input available to implement the proposed strategy, which is aimed to extend the operating limits of ω_r and τ_L of the SPh-BESM, ensuring safe commutation.

2) EFFECT OF α ON $I'_{FD_{EXC}}$ AND I_S

For the design of the proposed strategy, it is important to first understand the effect that α has on $i'_{fd_{\mathrm{EXC}}}$ and I_s. For this, according to (8), to maintain the stator flux magnitude Ψ_s constant, i'_{fd} must be directly controlled through $i'_{fd_{EXC}}$ on the SPh-BE side to compensate for changes in i_{Ts} and i_{Ms}. Therefore, i'_{fd} and $i'_{fd_{EXC}}$ depend on i_{Ts} and i_{Ms}. This dependency is described by the following current relations:

$$i_{Ms} = -i_{Ts} \tan \varphi, \tag{41}$$

$$I_s = \frac{i_{Ts}}{\cos \varphi}, \tag{42}$$

where the negative sign has been added because i_{Ms} is always negative as a result of the leading power factor that allows natural commutation. From (21), $\varphi = \arccos(-\cos(\alpha + \mu/2))$, that is, φ is a function of α and thus, from (42), I_s also depends on α.

Substituting (41) into (8) and solving for i'_{fd} yields

$$i'_{fd} = \frac{1}{L_{md}\cos\delta}\Big[\Psi_s + (L_{ls} + L_{md}\cos^2\delta + \\ + L_{mq}\sin^2\delta)i_{Ts}\tan\varphi - \left(\frac{L_{mq}-L_{md}}{2}\sin 2\delta\right)i_{Ts}\Big] \quad (43)$$

where Ψ_s remains constant, while i_{Ts} varies with the load torque.

Notice, from (43), that φ is a function of α, and therefore i'_{fd} also depends on α. Moreover, since $i'_{fd_{EXC}}$ is a function of i'_{fd}, $i'_{fd_{EXC}}$ also depends on α. Consequently, I_s, i'_{fd}, and $i'_{fd_{EXC}}$ can be modified through α as long as the safe commutation condition $\gamma/\omega_r \geq t_q$ (presented in (20)) is satisfied all times.

In what follows, an expression is obtained to calculate α from (18). This yields an expression in terms of i_{DC} and V_{LL}, which depend in their turn on φ. To eliminate this dependence on φ, it is proposed here to use the following descriptions:

$$i_{\mathrm{DC}} = \frac{\pi}{2\sqrt{3}}I_s, \quad (44)$$

$$V_{\mathrm{LL}} = \frac{v_{Ts}}{\cos\varphi}. \quad (45)$$

Direct substitution of (41), (44) and (45) in (18) yields the following expression for γ independent of φ:

$$\gamma = \arccos\left(\frac{\pi}{\sqrt{3}}\frac{\omega_r L_c i_{Ts}}{v_{Ts}} - \cos\alpha\right). \quad (46)$$

Notice that this expression is in terms of the T-component variables i_{Ts} and v_{Ts}, which are determined solely by the load torque.

Now that the expression of the margin angle γ is defined in terms of T components, the inequality (20) to ensure a safe commutation can be rewritten as

$$t_q\omega_r < \arccos\left(\frac{\pi}{\sqrt{3}}\frac{\omega_r L_c i_{Ts}}{v_{Ts}} - \cos\alpha\right). \quad (47)$$

During normal operation, the machine may be under unexpected oscillations and harmonics that may lead to a commutation failure if $\gamma = t_q\omega_r$. To prevent this situation, it is proposed to add an extinction time offset t_0 to t_q, that is,

$$\gamma = (t_q + t_0)\omega_r. \quad (48)$$

The extinction time offset t_0 imposes a safety margin that limits the commutation point (the value that α takes) to avoid the commutation failure area.

Finally, the following strategy is proposed for the α calculation:

$$\alpha = \arccos\left(\frac{\pi}{\sqrt{3}}\frac{\omega_r L_c i_{Ts}}{v_{Ts}} - \cos(\omega_r(t_q + t_0))\right), \quad (49)$$

whose block diagram is shown in Fig. 4.

Notice that, in equations (46), (47) and (49); the reactive commutation term remains at $\frac{\pi}{\sqrt{3}}\frac{\omega_r L_c i_{Ts}}{v_{Ts}}$, which is invariant regardless of the unit system used (SI or pu). However, the margin angle term $\cos(\omega_r(t_q + t_0))$ must be maintained in SI units. Therefore, if variables are measured in per unit, they should be scaled with the appropriate bases to ensure consistency.

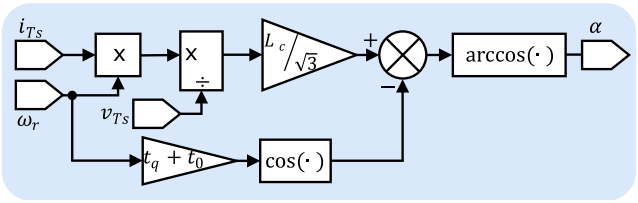


FIGURE 4. Block diagram for the computation of α according to (49).

To address practical DSP implementation and safeguard the analytical calculation against boundary conditions, the following protective measures were established. To ensure robust implementation under boundary conditions, the analytical calculation is bypassed in favor of forced-commutation pulsating at low speeds ($\omega_r < 5 - 10\%$ of nominal speed). Furthermore, during severe load transients, the inverse cosine argument is strictly clamped within $[-1, 0]$ to prevent numerical exceptions and avoid entering to rectifier mode, triggering an active reduction of the i_{Ts} reference if saturation occurs. To prevent abrupt commutation stress during transients, the rate of change of the computed firing angle is constrained by a firmware limiter ($d\alpha/dt$) that mitigates excessive dv/dt and di/dt stress. Additionally, if signal validation detects a sensor dropout, the controller immediately reverts to a safe and conservative predefined fallback angle, and initiates a protective trip sequence.

Finally, to address the control bandwidth and overall robustness of the system, the dynamic stability of the proposed approach is studied as follows. Dynamic stability is inherently guaranteed since the α calculation operates as an algebraic feedforward equation, avoiding closed-loop feedback instabilities. To prevent noise-induced oscillations near the commutation boundary, the firmware applies low-pass filtering, rate limiting ($d\alpha/dt$), and an operating point shift via the safety margin time ($t_q + t_0$). Ultimately, the practical adaptation bandwidth is constrained by the discrete firing frequency of the LCI, signal tracking delays, and programmed rate limiters that protect the thyristors.

D. TORQUE CONTROL

The α-modified FOC regulates the SM electromechanical torque τ_e through both the conventional speed and current controllers. Based on the conventional structure presented above, the torque control comprises the speed controller (23), and the DC-link current controller (24)-(26), and its block diagram is shown in Fig. 5.

E. STATOR FLUX MAGNITUDE CONTROL

The proposed stator flux magnitude control (part of the α-modified FOC) consists of a modification to the conventional controller (presented in Section III-B) conformed by the flux controller (28) and (30), and the field current controller (31)-(32). The proposed modification includes the

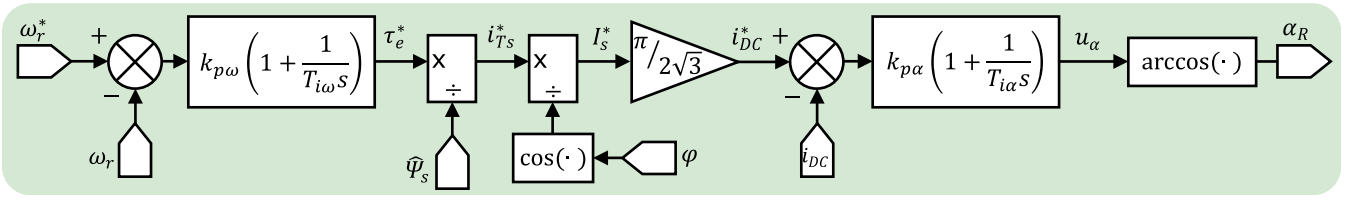


FIGURE 5. Block diagram of the conventional torque control in terms of *MT*-coordinates.

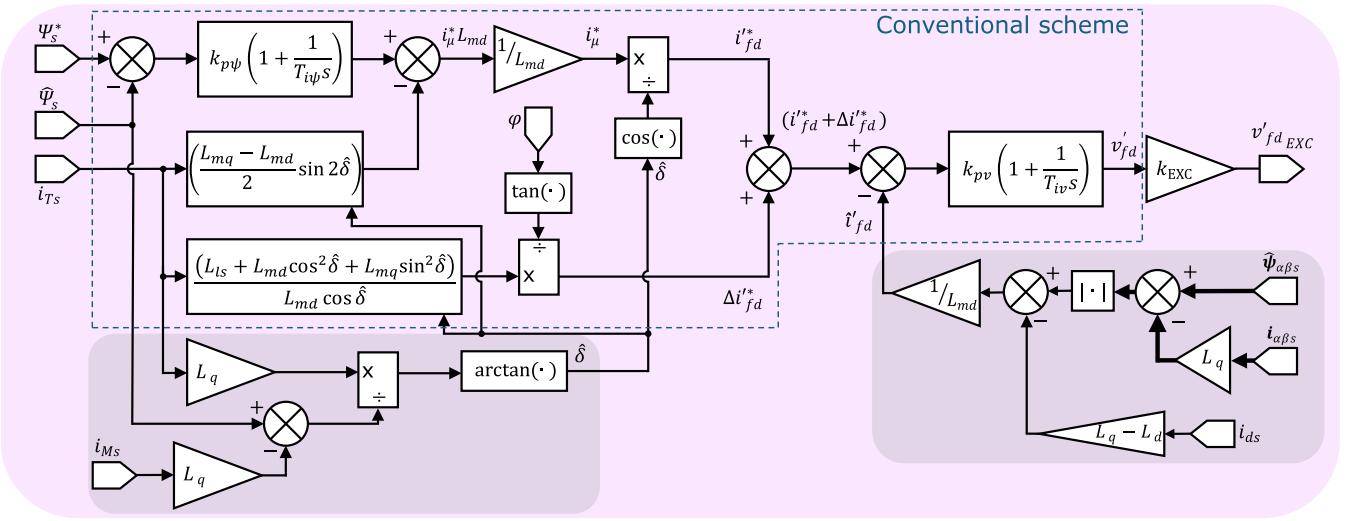


FIGURE 6. Block diagram of the proposed stator flux magnitude control based on the conventional scheme with added modifications: **(bottom left-hand side)** torque angle estimation, and **(bottom right-hand side)** SM field current estimation.

estimation of $\hat{\delta}$ (33), the estimation of $\hat{i}'_{fd}$ (36), and the consideration of the model simplification of the SPh-BE represented by constant K_{BE}(38). Fig. 6 shows the block diagram of the proposed stator flux magnitude controller.

Summarizing, the proposed controller consists of an alternative load synchronized switching policy b, Table 2; the consideration of the simplified model of the SPh-BE, (38); an inverter firing angle controller (49) and the addition of an adaptive flux observer, all of them integrated with a torque and stator flux controls.

V. EXPERIMENTAL RESULTS

This section details experimental results using hardware in the loop (HIL) setup to assess the proposed α-modified FOC scheme for controlling a 100 kW LCI-based drive SPh-BESM. Real-time simulation (RTS) was implemented on a dSpace 1007 platform, where the complete control algorithm is executed at a fixed sampling rate of 278 μs, resulting in a real-time computational burden of approximately 70% of the CPU capacity. Since the proposed adaptation relies on a direct algebraic calculation rather than computationally heavy iterative solvers or multi-dimensional interpolations, it is perfectly feasible to implement it in standard industrial DSP or FPGA-based control platforms.

Simulation uses a mode-switching sequence governed by a five-state logic to ensure a stable start-up and controller handover. The sequence begins with an asynchronous start via damper windings with an active crowbar (State 1) until rotor speed exceeds 0.95 pu. The crowbar is then automatically deactivated and field forcing is applied to achieve grid synchronism (State 2). Once pull-in stability and a leading power factor are confirmed, the LCI is initialized in tracking mode (State 3), where the control PI integrators are accurately preset using real-time feedback to match initial firing angles with the machine back-EMF. After opening the main AC breaker and verifying zero stator current following a mandatory dead-time, seamless handover is executed (State 4) by unblocking the LCI pulses to minimize torque dip, which finally goes into full dynamic control (State 5). If synchronization conditions are not met within a predefined window, the protection system aborts the sequence and trips the main breaker.

It is worth noting that, while the current HIL setup validates real-time computational feasibility, a full-scale physical deployment would require standard industrial sensors (DC-link current, grid/machine voltages and currents, exciter feedback, and a 1024 PPR encoder) alongside baseline machine parameters (resistance and inductances). To manage the computational load in a real-world scenario, a DSP or advanced microcontroller would need to execute the main FOC loop at a minimum frequency of 3.6 kHz. Furthermore, for industrial robustness, delegating the switching scheme

to an FPGA would ensure high-accuracy triggering, while fiber-optic gate drives would provide the essential noise immunity.

A. ROTOR SPEED AND ESTIMATED STATOR FLUX MAGNITUDE REGULATION

Rotor speed and load torque profiles were designed to emulate a milling process, that is, ω_r^* decreases from 1 pu to 0.8 pu, and simultaneously τ_L grows from 1 pu to 1.25 pu.

As shown in Fig. 7, the controller forces both ω_r and τ_e to follow the speed profile reference. Figure 8 shows that $\hat{\Psi}_s$ tracks its reference generated by the controller itself. To implement the flux weakening strategy, the stator flux magnitude reference Ψ_s^* is maintained at 1 pu for $\omega_r \leq 1$ pu, while, for $\omega_r > 1$ pu, the controller reduces Ψ_s^* to a value inversely proportional to ω_r.

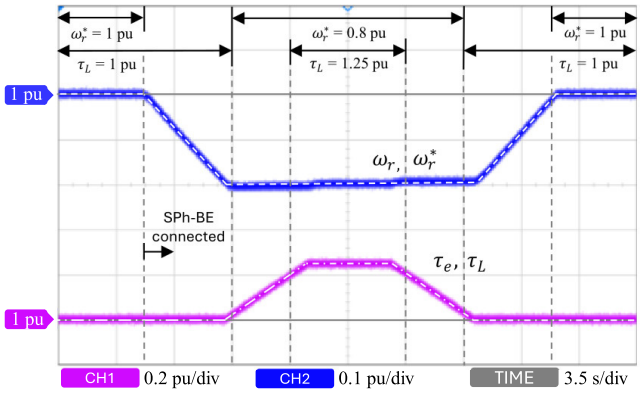


FIGURE 7. Time response under α-modified FOC. In this test, ω_r^* fluctuates between 1 pu and 0.8 pu and τ_L fluctuates between 1 pu and 1.25 pu. The SPh-BE is connected at $t = 5$ s. (top plots) Rotor speed ω_r (solid) and its reference ω_r^* (dashed); (bottom plots) the electromagnetic torque τ_e (solid) and its reference τ_L (dash-dotted).

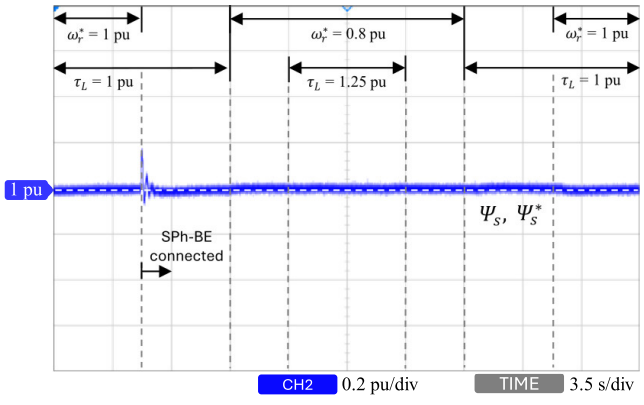


FIGURE 8. Time response under α-modified FOC. In this test, ω_r^* fluctuates between 1 pu and 0.8 pu and τ_L fluctuates between 1 pu and 1.25 pu. The SPh-BE is connected at $t = 5$ s. Estimated stator flux magnitude $\hat{\Psi}_s$ (solid) and its reference $\hat{\Psi}_s^*$ (dashed).

To validate the drive's dynamic performance across a broader operating range, the system was evaluated under a wider conditions range. The control successfully handled simultaneous transitions between a low-speed/high-torque region (0.8 pu speed, 1.15 pu load) and a high-speed flux-weakening region (1.1 pu speed, 0.85 pu load) without losing stability. Furthermore, the dynamic tracking performance indices were quantified as described next. A speed step from 1.0 pu to 1.1 pu yielded a peak overshoot of 1.126 pu, a settling time of 650 ms, and zero steady-state error in average. Similarly, a torque reference step from 0.9 pu to 1.0 pu resulted in a peak overshoot of 1.07 pu, a settling time of 400 ms, and zero steady-state error in average. These indices confirm the robust regulation and fast-tracking capabilities of the proposed control scheme under several industrial conditions.

It is worth mentioning that the flux observer was digitally implemented using the backward Euler approximation at a 3.6 kHz sampling rate. To reduce noise, the low-pass filters were tuned at a 6 Hz cutoff frequency, while the PI adaptation mechanism used an integral gain of 0.08 rad/s. To study the flux observer sensitivity to parameter mismatch and noise, its performance was evaluated by deliberately introducing increments to the stator resistance parameter (R_s) across a $\pm 20\%$ range. The results demonstrated excellent robustness even under the worst-case of 20% mismatch caused by thermal drift or a defective measurement. The error in the estimated stator flux magnitude remained as low as 1.1% as seen in 3.

TABLE 3. Parameter sensitivity of the stator flux observer to stator resistance variations.

Stator resistance (R_s)	R_s error [%]	Estimation of stator flux magnitude [pu]	Flux estimation error [%]
$1.2R_s$	20%	0.468	-1.12%
$1.1R_s$	10%	0.4706	-0.57%
R_s (nom)	0%	0.4733	0.00%
$0.9R_s$	-10%	0.4759	0.55%
$0.8R_s$	-20%	0.4785	1.10%

Beyond the reduced impact on amplitude, the results reveal a crucial operational advantage: while the magnitude of the flux varies slightly under variations of R_s, the phase angle and zero crossings remain completely unaffected. Since the thyristor firing logic relies strictly on these crossings for synchronization, the proposed scheme remains inherently robust against resistance mismatch and thermal drift.

To evaluate the cascading effect of parameter mismatch on the overall drive stability, the impact of estimation errors on the firing angle was quantified as described next. A worst-case 1.12% flux estimation error induces a 1.13% compensatory increase in the stator current reference to maintain the load torque. Since this current adjustment only scales the relatively small commutation overlap term, the final evaluated α shifts by less than 0.2%. This negligible deviation demonstrates that the scheme inherently absorbs thermal parameter variations, preserving precise and safe commutation margins without risk of an inversion failure.

Finally, inductance mismatches in the feedforward terms are naturally compensated by the current PI regulators. However, the analytical load angle (δ) is highly sensitive

to parameter variations: a ±20% error in L_q shifts δ by 7°, inducing a cross-coupling between the torque and flux axes. To completely eliminate this parameter dependence, the drive may utilize a direct geometric decoupling, defined as $\delta = \mu - \theta_r$. By calculating δ from the estimated stator flux angle and the measured rotor position, the MT-axis decoupling is rendered entirely immune to inductance mismatch.

B. BENCHMARK COMPARISON: α-MODIFIED FOC VS. FIXED-FIRING ANGLE

A comparison between the proposed α-modified FOC and the fixed-firing angle conventional FOC scheme is presented using the same ω_r^* and τ_L profiles. For the latter, a constant $\alpha = 2.27$ rad has been considered, where the selected value guarantees safe commutation during the process. Figure 9 shows the α signals of both schemes.

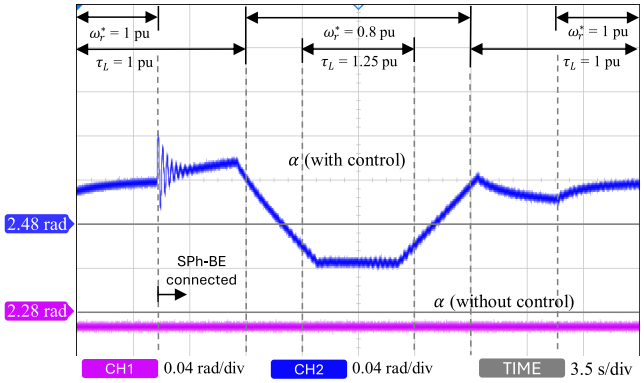


FIGURE 9. Time response of the comparison between α-modified FOC and the conventional FOC scheme. In this test, ω_r^* fluctuates between 1 pu and 0.8 pu and τ_L fluctuates between 1 pu and 1.25 pu. The SPh-BE is connected at $t = 5$ s. Controlled α calculated online under the proposed α-modified FOC scheme (top), and constant $\alpha = 2.27$ rad under the conventional FOC scheme (bottom).

Figure 10 shows $i'_{fd_{\text{EXC}}}$, which is a variable that restricts the operating range of ω_r^* and τ_L in SPh-BESM. Notice that a *controlled* α obtained with the proposed α-modified scheme results in a lower current as compared to the fixed-angle case, where a *constant* α is considered. It is observed that $i'_{fd_{\text{EXC}}}$ is reduced by approximately 20% on average.

As shown in Fig. 11, it is observed that I_s was also reduced by 17% on average under the proposed α-modified scheme.

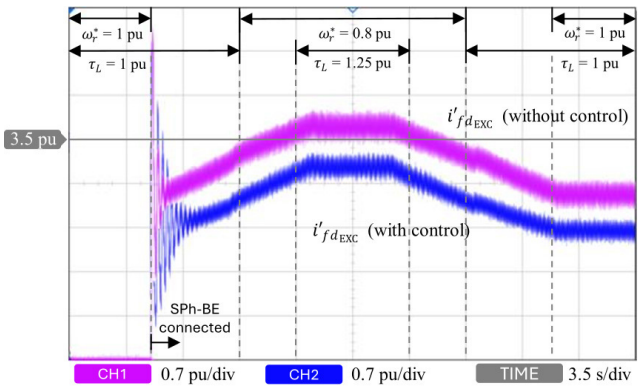


FIGURE 10. Time response of the comparison between α-modified FOC and the conventional FOC scheme. In this test, ω_r^* fluctuates between 1 pu and 0.8 pu and τ_L fluctuates between 1 pu and 1.25 pu. The SPh-BE is connected at $t = 5$ s. $i'_{fd_{\text{EXC}}}$ with α calculated online with the proposed α-modified FOC scheme (top) and $i'_{fd_{\text{EXC}}}$ with constant α under conventional FOC scheme (bottom).

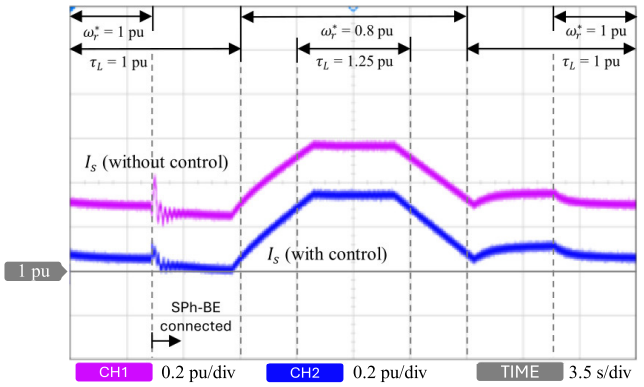


FIGURE 11. Time response of the comparison between α-modified FOC and the conventional FOC scheme. In this test, ω_r^* fluctuates between 1 pu and 0.8 pu and τ_L fluctuates between 1 pu and 1.25 pu. The SPh-BE is connected at $t = 5$ s. I_s with α calculated online with the proposed α-modified FOC scheme (top) and I_s with constant α under conventional FOC scheme (bottom).

To quantify the expansion of the operating range achieved with the proposed α-modified FOC, ω_r^* is decreased and τ_L is increased, one at a time, while α is calculated online and the currents reach their maximum values observed under the fixed-angle conventional scheme. This yields the following two test conditions:

(i) τ_L=1.25 pu, ω_r^* is gradually reduced to 0.65 pu, which corresponds to a maximum $i'_{fd_{\text{EXC}}}$=3.75 pu. This represents a reduction of 18.75% in ω_r^* as shown in Fig. 12.

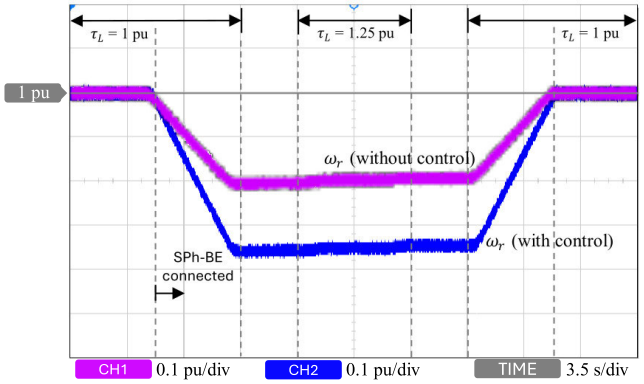


FIGURE 12. Time response of the comparison between α-modified FOC and the conventional FOC scheme. In this test, ω_r^* is gradually reduced to 0.65 pu, which corresponds to a maximum $i'_{fd_{\text{EXC}}}$=3.75 pu. The SPh-BE is connected at $t = 5$ s. ω_r with α calculated online with the proposed α-modified FOC scheme (top) and ω_r with constant α under conventional FOC scheme (bottom).

(ii) ω_r^*=0.8 pu, τ_L is gradually increased to 1.45 pu, which corresponds to a maximum I_s=1.58 pu. This represents an increment of 13.8% in τ_L, as shown in Fig. 13.

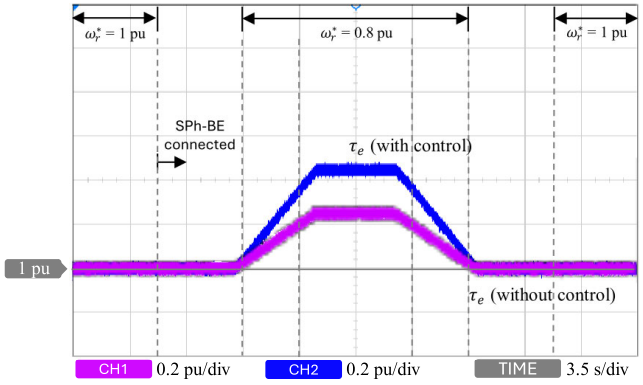


FIGURE 13. Time response of the comparison between α-modified FOC and the conventional FOC scheme. In this test, τ_L is gradually increased to 1.45 pu, which corresponds to a maximum I_s=1.58 pu. The SPh-BE is connected at $t = 5$ s. τ_e with α calculated online with the proposed α-modified FOC scheme (top) and τ_e with constant α under conventional FOC scheme (bottom).

Strictly limiting the upper current limits to the maximum values observed under the conventional fixed-angle scheme establishes an equal comparative voltage baseline. This condition guarantees that the demonstrated operating range expansions are achieved purely by reclaiming reactive power capacity through dynamic angle optimization, imposing zero additional thermal or electrical stress on the drive components.

Remark 2: The maximum limits of $i'_{fd_{\text{EXC}}}$ and I_s have been selected as the maximum values observed under the fixed-angle conventional scheme, that is, when α is kept constant. In this case, $i'_{fd_{\text{EXC}}} = 3.75$ pu and $I_s = 1.58$ pu.

Furthermore, the dynamic optimization of α inherently improves the system efficiency. In LCI drives, the total copper losses (P_{cu}) across the DC-link reactor (R_{DC}) and stator windings (R_s), for a given active power P, are rigorously defined by incorporating the fundamental power factor angle, i.e.,

$$P_{cu}(\alpha) = \frac{P^2}{V_{LL}^2 \cos\left(\alpha + \frac{\mu}{2}\right)^2}\left(\frac{R_{DC}}{2} + R_s\right). \tag{50}$$

By dynamically optimizing α to its safe maximum, the proposed scheme maximizes the power factor; therefore, minimizing system currents and conduction losses compared to fixed-margin methods.

C. BENCHMARK COMPARISON: α-MODIFIED FOC VS. 2D LOOK-UP TABLE

A comparison between the proposed analytical α-calculation and a conventional 2D LUT is presented using the same ω_r^* and τ_L profiles. The LUT shown in Table 4 was populated offline to guarantee a 15° commutation margin. However, comparative results revealed a critical flaw in the LUT approach: its construction systematically underestimates the required firing angle due to the highly non-linear nature of the exact arccosine solution. While this prevents commutation failure, it forces an excessively large margin, degrading the displacement power factor and drawing unnecessary reactive power. In contrast, the proposed algebraic method continuously tracks the exact optimal α, ensuring safe commutation while maximizing system efficiency.

TABLE 4. 2-D look-up table for firing angle α (rad).

	i_{Ts} (pu)				
ω_r (pu)	**0.8**	**0.9**	**1.0**	**1.1**	**1.2**
0.8	2.4686	2.4302	2.3934	2.3579	2.3239
0.9	2.4213	2.3804	2.3413	2.3035	2.2670
1.0	2.3756	2.3323	2.2907	2.2508	2.2118
1.1	2.3309	2.2853	2.2415	2.1993	2.1581
1.2	2.2872	2.2394	2.1935	2.1488	2.1054

Figure 14 shows the α signals of both schemes.

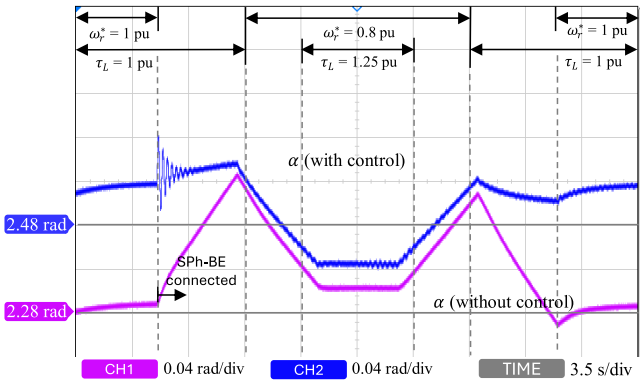


FIGURE 14. Time response of the comparison between α-modified FOC and the LUT-based conventional FOC scheme. In this test, ω_r^* fluctuates between 1 pu and 0.8 pu and τ_L fluctuates between 1 pu and 1.25 pu. The SPh-BE is connected at $t = 5$ s. Controlled α calculated online under the proposed α-modified FOC scheme (top), and LUT-calculated α rad under the conventional FOC scheme (bottom).

Figure 15 shows $i'_{fd_{\text{EXC}}}$, which is a variable that restricts the operating range of ω_r^* and τ_L in SPh-BESM. Notice that a *controlled* α obtained with the proposed α-modified scheme results in a lower current as compared to case, where a *LUT-calculated* α is considered. It is observed that $i'_{fd_{\text{EXC}}}$ is reduced by approximately 16.55% on average.

As shown in Fig. 11, it is observed that I_s was also reduced by 13.5% on average under the proposed α-modified scheme.

To quantify the expansion of the operating range achieved with the proposed α-modified FOC as compared to the LUT-based α, ω_r^* is decreased and τ_L is increased, one at a time, while α is calculated online until the currents reach their maximum values observed under the LUT-based conventional scheme.

(i) τ_L=1.25 pu, ω_r^* is gradually reduced to 0.67 pu, which corresponds to a maximum $i'_{fd_{\text{EXC}}}$=3.62 pu. This represents a reduction of 16.25% in ω_r^* as shown in Fig. 12.

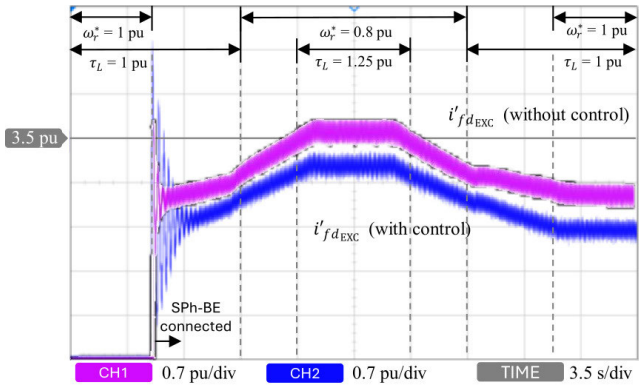


FIGURE 15. Time response of the comparison between α-modified FOC and the LUT-based conventional FOC scheme. In this test, ω_r^* fluctuates between 1 pu and 0.8 pu and τ_L fluctuates between 1 pu and 1.25 pu. The SPh-BE is connected at $t = 5$ s. $i'_{fd_{EXC}}$ with α calculated online with the proposed α-modified FOC scheme (top) and $i'_{fd_{EXC}}$ with LUT-calculated α under conventional FOC scheme (bottom).

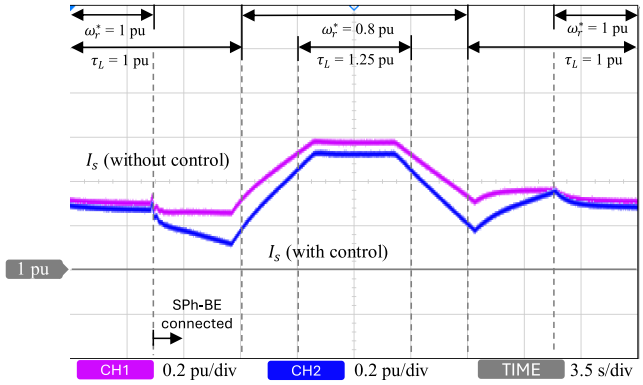


FIGURE 16. Time response of the comparison between α-modified FOC and the LUT-based conventional FOC scheme. In this test, ω_r^* fluctuates between 1 pu and 0.8 pu and τ_L fluctuates between 1 pu and 1.25 pu. The SPh-BE is connected at $t = 5$ s. I_s with α calculated online with the proposed α-modified FOC scheme (top) and I_s with LUT-calculated α under conventional FOC scheme (bottom).

(ii) ω_r^*=0.8 pu, τ_L is gradually increased to 1.4 pu, which corresponds to a maximum I_s=1.54 pu. This represents an increment of 12% in τ_L, as shown in Fig. 13.

D. LOAD-SYNCHRONIZED SWITCHING POLICY

Figure 17 shows $\hat{\boldsymbol{\psi}}_{abcs}$ and -$\hat{\boldsymbol{\psi}}_{abcs}$ obtained from the adaptive flux observer, and two pulses synchronized, T_{I1} and T_{I3}. The vertical dotted line marks the natural commutation point of the thyristor where α is measure from. As compared to the actual line-to-line voltages measured in the SM terminals shown in Fig. 2, the *abc*-estimated stator fluxes are smooth signals that improve the pulse generation accuracy.

To quantify the performance of pulse generation, both schemes voltage synchronization and flux synchronization, were evaluated under heavy-load transient conditions. Deep commutation notches caused conventional voltage-based tracking to exhibit a severe 1.286 ms (27.8°) phase error. This

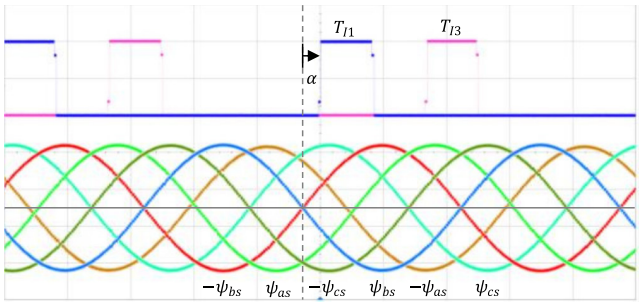


FIGURE 17. *abc*-estimated stator fluxes (bottom). The vertical dotted line marks the natural commutation point of the thyristor, serving as the start point of α before T_{I1} can be produced (top).

load-dependent delay consumes the safe commutation margin and cannot be effectively compensated with a static offset, even though it does not alter the current THD. In contrast, the stator flux inherently filters out these notches. The proposed flux-based synchronization logic in 2 entirely eliminates this 27.8° timing error by relying on clean zero-crossings, ensuring highly precise pulse generation that enables the safe optimization of α near its theoretical limits.

VI. CONCLUSION

This paper presented the design and analysis of the proposed α-modified FOC. The proposed scheme aimed to control an LCI-based drive for a SPh-BDSM. HIL experiments, considering the effect of the SPh-BE, showed that both rotor speed and stator flux magnitude controllers met their control objectives guaranteeing regulation towards a milling process profile.

The proposed α-modified FOC incorporated a scheme for online calculation of the firing angle α, with the benefit of extending the operating ranges of rotor speed and load torque as compared to either the conventional FOC scheme producing a fixed-angle and the LUT approach generating a sequence of angles.

Unlike conservative baseline strategies, the proposed analytical formulation established an absolute safe operating limit that continuously optimizes the commutation angle. This dynamic calculation minimizes the required capacitive reactive power, thereby reducing the overall magnitude of the stator current, copper losses, and thermal stress. Furthermore, by coupling the firing angle directly to instantaneous load dynamics, the system ensures fast feedforward adaptation during severe transients, magnetic saturation, and field weakening, preventing margin collapse and mitigating excessive dv/dt and di/dt stress.

Hardware-in-the-loop tests showed that the rotor speed is reduced up to 18.75%, and the load torque is augmented by 13.8% thanks to the use of α-modified FOC instead of the inverter fixed-firing angle. Additionally, tests showed a reduction of the excitation field current by 20% and the stator current by 17% on average, which is the mechanism that allows the expansion of the operating range.

It also showed that the rotor speed is reduced up to 16.25%, and the load torque is augmented by 12% when compared to LUT-based α. Additionally, it showed a reduction of the excitation field current by 16.55% and the stator current by 13.5% on average.

The proposed α-modified FOC also introduces an enhanced switching policy that produces smoother pulses compared to conventional methods. This policy is synchronized with the load through estimated stator flux *abc*-signals free from commutation peaks and notches. Deep commutation notches cause conventional voltage-based tracking to exhibit a severe 1.286 ms (27.8°) phase error. The proposed flux-based synchronization logic in 2 entirely eliminates this 27.8° timing error by relying on clean zero-crossings, ensuring highly precise pulse generation that enables the safe optimization of α near its theoretical limits.

Regarding the scalability, since LCI technology is inherently reserved for multi-megawatt drives, the proposed control scheme operates directly within its intended high-power environment. Applying this methodology to even higher-rated machines requires no modifications to the core control logic nor introduces additional computational complexity.

While highly effective for high-power industrial drives, the practical implementation of the proposed scheme is currently limited by its reliance on a mechanical encoder and its sensitivity to machine parameter variations caused by thermal drift and magnetic saturation. Future work will address these constraints by developing sensorless control observers to eliminate the encoder, implementing online parameter estimation for continuous self-adaptation, and transitioning from HIL testing to experimental validation on a scaled LCI drive with a brushless exciter.

APPENDIX
SYSTEM PARAMETERS

See Tables 5 and 6.

TABLE 5. Machine parameters.

Description	Symbol	Value	Unit
Stator resistance	R_s	0.02599	pu
Stator leakage inductance	L_l	0.07	pu
d-axis mutual inductance	L_{md}	2.25	pu
q-axis mutual inductance	L_{mq}	0.99	pu
Field winding resistance	R_{fd}	0.016	pu
Field winding leakage inductance	L_{lfd}	0.2092	pu
d-axis damper resistance	R_{kd}	0.2047	pu
d-axis damper leakage inductance	L_{lkd}	0.6524	pu
q-axis damper resistance	R_{kq}	0.04286	pu
q-axis damper leakage inductance	L_{lkq}	0.06387	pu
Inertia constant	H	0.1268	s
Pole pairs	p	2	-

TABLE 6. Base values and LCI parameters.

Description	Symbol	Value	Unit
Base Values			
Base voltage	V_b	375	V
Base power	P_b	100	kVA
Base current	I_b	180	A
Base impedance	Z_b	2.11	Ω
Base inductance	L_b	5.60	mH
Base angular frequency	ω_b	377	rad/s
LCI Parameters			
Commutation inductance	L_c	0.948	mH
DC link inductance	L_{DC}	400	μH
Source frequency	f_{source}	60	Hz
Source voltage	V_{source}	460	V

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